

PRINCE CHAUHAN

Senior Quantitative Researcher | Options, Volatility & Systematic Trading

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github.com/prince-chauhan-pc/quant-research-portfolio | SEBI Registered Research Analyst

PROFESSIONAL SUMMARY

Derivatives research professional with 9+ years of experience spanning options trading, market analysis, and systematic strategy development. Combines deep institutional derivatives expertise with AI-augmented quantitative engineering to design, prototype, and rigorously validate production alpha and execution pipelines across Indian equity derivatives and US index options. Experienced in production alpha validation, multi-crore SPAN margin allocation, and automated research architectures.

PROFESSIONAL EXPERIENCE

SMC Global Securities Ltd. - Senior Quantitative Analyst & Researcher

Dec 2024 - Present

- **Alpha Research Infrastructure:** Architected causal event-driven simulation pipelines and point-in-time alpha research on tick-level and 1-minute historical bar feeds across US and Indian index derivatives (SPX, VIX, Nifty, Bank Nifty).
- **US Index Derivatives:** Engineered automated execution pipelines and volatility surface modeling for short-dated US index options (SPX/SPXW) via Interactive Brokers (IBKR API), systematically capturing short-horizon IV skew mispricings with dynamic delta rebalancing.
- **Statistical Arbitrage:** Traded mean-reverting equity basket spreads across Bank Nifty constituents, maintaining strict delta neutrality via dynamic beta hedging and automated strike re-centering thresholds.
- **Walk-Forward Strategy Validation:** Delivered 2.5+ out-of-sample Sharpe and 2.1+ Profit Factor across a 7-year sample (2018-2024) of Nifty & Bank Nifty index options, evaluated net of 0.50% turnover friction.
- **TCA & Execution Optimization:** Reduced execution drag by 24% and eliminated broker order-state desynchronization via pre/post-trade Transaction Cost Analysis (TCA) telemetry and unique client order tracking.
- **Desk Mentorship & Risk Alignment:** Mentored 3 junior quantitative researchers and traders in statistical validation methods and execution best practices; partnered with Portfolio Managers on strategy scalability, real-time portfolio Greeks, dealer GEX mapping, parametric VaR, and SPAN margin caps.

Smart Bulls Capital - Research Analyst, Derivatives & Quantitative Research

Mar 2017 - Nov 2024

- **Options Volatility Books:** Researched, structured, and traded systematic multi-leg options volatility strategies (Iron Condors, Butterflies, Calendar Spreads) with dynamic delta/vega rebalancing guided by real-time IV surface spline smoothing.
- **Implied Volatility & Greeks:** Inverted Black-Scholes prices numerically via Newton-Raphson and Brent algorithms; mapped dealer Gamma Exposure (GEX) to identify institutional pin-risk levels across weekly expiries.
- **Portfolio Construction & Allocation:** Implemented Risk Parity and Mean-Variance capital allocation models across multi-strike options books; managed daily P&L attribution, SPAN margin utilization, and drawdown limits within desk risk budgets.
- **Live Capital & Margin Scale:** Managed systematic intraday and multi-day options books deploying multi-crore margin; maintained strict SPAN margin utilization and contained drawdowns within desk risk tolerances.

Earlier Corporate Experience - Corporate Finance & Audit Roles

2014 - 2017

- **Corporate Treasury & Audit:** Executed treasury operations and working-capital forecasting at Indian Oil Corporation Ltd. (IOCL, 2016-2017); conducted statutory audits and corporate tax structuring at CA Matta & Associates (2014-2016).

QUANTITATIVE RESEARCH ARCHITECTURE & EMPIRICAL FRAMEWORKS

Hosted at: github.com/prince-chauhan-pc/quant-research-portfolio

Point-in-Time Backtesting Framework & Invariant Gate

- **Research Objective:** Enforce point-in-time data alignment and realistic transaction friction in high-turnover options strategy research.
- **Methodology & Validation:** Implemented rolling walk-forward optimization partitioned into 70% in-sample parameter calibration, 20% out-of-sample cross-validation, and 10% held-out test windows to prevent data leakage, paired with 1,000-path Monte Carlo drawdown stress testing.
- **Audit Guardrails:** Engineered an automated data-integrity test suite enforcing Rules L1-L17 across 22 boundary checks, pre-deducting 0.50% round-trip premium friction and exchange SPAN margins.

Arbitrage-Free SVI Volatility Surface & GEX Profiling

- **Research Objective:** Eliminate calendar and butterfly arbitrage from market implied volatility quotes for robust cross-strike option pricing.
- **Methodology:** Calibrated daily implied volatility surfaces using SVI parameterization with non-linear optimization; enforced butterfly convexity and calendar-spread non-arbitrage constraints.
- **Practical Outcome:** Stabilized cross-strike smile interpolation across illiquid strikes and generated real-time dealer GEX sensitivity curves to map institutional pin-risk zones.

Bank Nifty Basket Cointegration & Statistical Arbitrage

- **Research Objective:** Systematically exploit stationary mean-reverting spreads among top banking constituent equities.
- **Methodology & Risk:** Applied Johansen multivariate cointegration tests to identify stationary linear combinations; estimated spread mean-reversion half-life to calibrate entry/exit thresholds and holding periods, maintaining delta neutrality via rolling OLS beta weights.

Microstructure Execution Simulator & TCA Engine

- **Research Objective:** Accurately quantify market crossing costs and adverse selection for algorithmic limit orders.
- **Methodology:** Modeled Limit Order Book (LOB) queue priority, non-linear market impact, and fill probability across market regimes classified by Parkinson realized volatility.

TECHNICAL & QUANTITATIVE COMPETENCIES

- **Quantitative Research:** Statistical Arbitrage, Johansen Cointegration, Dynamic Beta Hedging, SVI Volatility Surface Fitting, Parkinson Realized Volatility, Market Regime Classification (K-Means), Walk-Forward Optimization, Monte Carlo Simulation.
- **Derivatives & Risk:** Black-Scholes Pricing, Portfolio Greeks (Delta, Gamma, Vega, Theta), Dealer GEX Profiling, Parametric & Historical VaR, SPAN Cross-Margin Optimization, Risk Parity, Mean-Variance Allocation.
- **AI-Augmented Engineering:** Agentic LLM Architecture for Rapid Alpha Prototyping, Automated Pipeline Construction, Synthetic Scenario Generation, Code Verification, and Mathematical Invariant Auditing.
- **Core Languages & Systems:** Python (NumPy, SciPy, Polars), SQL (DuckDB, PostgreSQL), Apache Parquet, Git, Linux.
- **Markets & Microstructure:** Indian Equity Derivatives (NSE Nifty, Bank Nifty, Sensex), US Equity Derivatives (CBOE SPX, VIX), Pre/Post-Trade TCA, Limit Order Book Queue Dynamics, Smart Order Slicing.

EDUCATION & REGULATORY CREDENTIALS

- **Bachelor of Commerce (Honours):** Kirori Mal College, University of Delhi (2011 - 2014) | First Division (71% Marks)
- **Chartered Accountancy (CA Final):** Institute of Chartered Accountants of India (ICAI, 2018) | Cleared Group 1 (78% Marks)
- **Regulatory Licenses:** SEBI Registered Research Analyst | NISM Series-XV